PROJECT MANAGEMENT

Assignment 4: Risk Management Plan and Risk Register

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1. Risk Governance Framework & Management Lifecycle

1.1 ISO 31000 & PMBOK 7th Edition Alignment

Risk management in ShalaConnect is governed by ISO 31000:2018 risk management guidelines and the PMBOK Guide 7th Edition risk performance domain. Risk is formally defined as an uncertain event or condition that, if it occurs, has a positive (opportunity) or negative (threat) effect on one or more project objectives (scope, schedule, cost, or quality).

1.2 The Five-Stage Risk Management Lifecycle

The project implements a continuous 5-stage lifecycle:
1. **Risk Identification:** Comprehensive elicitation using technical requirement checklists, architecture reviews, and stakeholder interviews across cluster schools.
2. **Qualitative Risk Analysis:** Prioritization of risks using standardized 5x5 Probability-Impact rating scales.
3. **Quantitative Risk Analysis:** Modeling monetary exposures and calculating the required financial contingency reserve.
4. **Risk Response Planning:** Engineering specific playbooks: Mitigate, Avoid, Transfer, or Accept for threats; Exploit, Enhance, Share, or Accept for opportunities.
5. **Risk Monitoring & Control:** Real-time monitoring of automated trigger conditions, daily standup reviews, and reserve drawdown logging.

1.3 Probability and Impact 5-Point Rating Scales

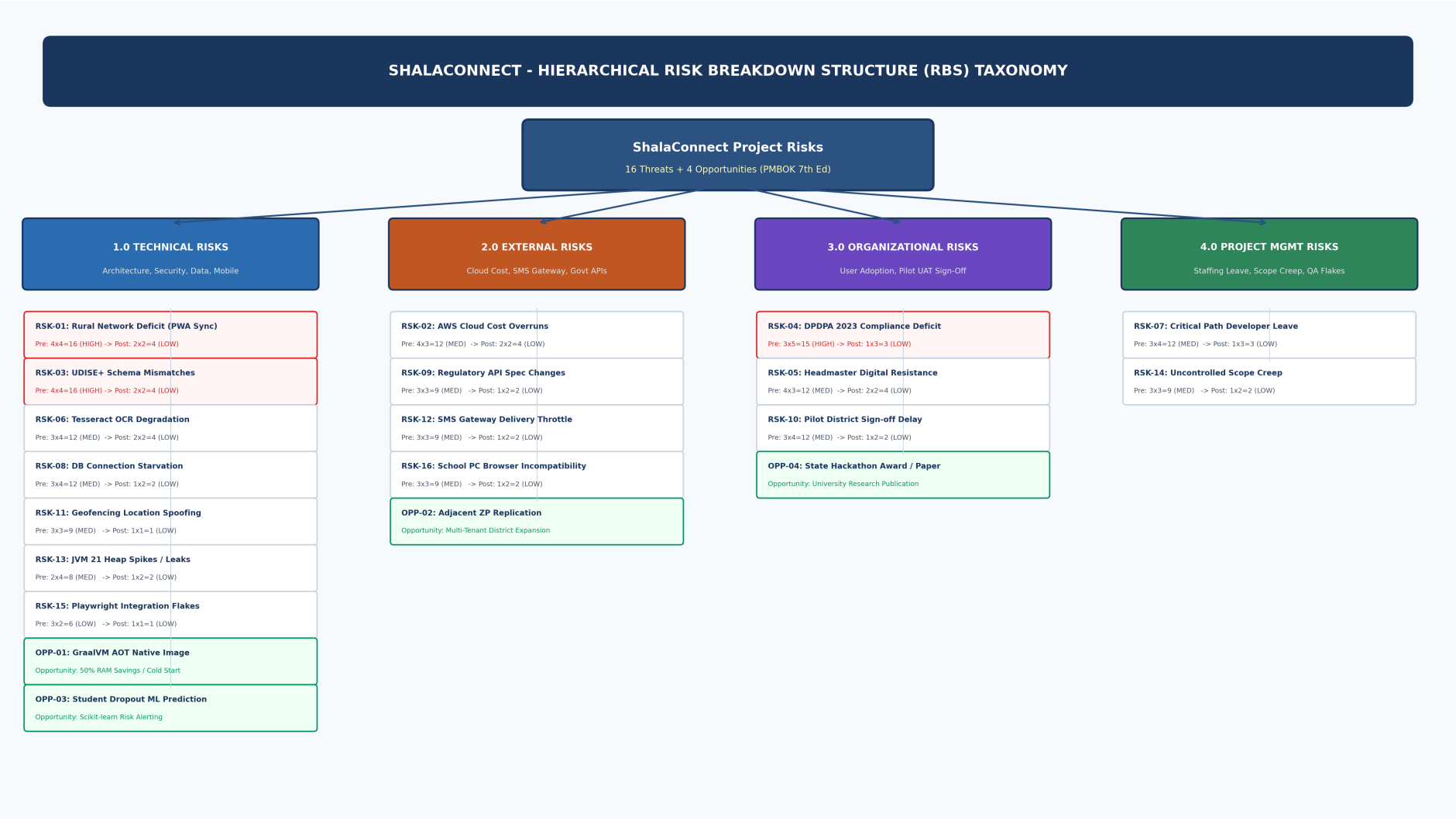
To ensure consistency, objective quantitative scales are defined across five rating tiers:
• **Probability (P):** 1 = Very Low (<10%), 2 = Low (10-30%), 3 = Medium (30-50%), 4 = High (50-70%), 5 = Very High (>70%).
• **Impact (I):**
  - *1 (Negligible):* Cost impact < INR 2,000; schedule delay < 1 day; negligible scope degradation.
  - *2 (Minor):* Cost impact INR 2,000 - 5,000; schedule delay 1-2 days; minor workaround available.
  - *3 (Moderate):* Cost impact INR 5,000 - 15,000; schedule delay 3-4 days; non-critical path affected.
  - *4 (Significant):* Cost impact INR 15,000 - 30,000; schedule delay 5-7 days; critical path threatened.
  - *5 (Catastrophic):* Cost impact > INR 30,000; schedule delay > 7 days; regulatory non-compliance or pilot cancellation.
• **Quantitative Risk Score:** $\text{Risk Score} = \text{Probability (P)} \times \text{Impact (I)}$.
• **Severity Thresholds:** **High** (Score 15-25, Red), **Medium** (Score 8-14, Amber), **Low** (Score 1-7, Green).

1.4 Expected Monetary Value (EMV) Modeling & Contingency Reserve Calculation

Expected Monetary Value (EMV) is calculated as $EMV = P \times I_{\text{financial}}$ across identified risks. Summing the EMVs for all 16 project threats produces a statistical financial exposure of **INR 38,600**. To provide a 95% statistical confidence buffer against concurrent risk triggers, the Project Sponsor formally authorized a **10% Contingency Reserve of INR 41,000** tied to the Cost Baseline (BAC = INR 4,10,000).

Threat ID	Risk Description	Prob % (P)	Est. Financial Impact (INR)	Calculated EMV (INR)	Allocated Reserve (INR)
RSK-01	Rural Network Connectivity Deficits	70% (0.7)	**INR 10,000**	**INR 7,000**	**INR 8,000**
RSK-02	AWS Cloud Infrastructure Cost Overruns	50% (0.5)	**INR 8,000**	**INR 4,000**	**INR 5,000**
RSK-03	UDISE+ Schema Ingestion Failures	70% (0.7)	**INR 8,000**	**INR 5,600**	**INR 6,000**
RSK-04	DPDPA 2023 Compliance Vulnerabilities	30% (0.3)	**INR 20,000**	**INR 6,000**	**INR 7,000**
RSK-05	Headmaster Adoption Resistance	50% (0.5)	**INR 6,000**	**INR 3,000**	**INR 4,000**
RSK-06	OCR Verification Accuracy Flaws	50% (0.5)	**INR 5,000**	**INR 2,500**	**INR 3,000**
RSK-07	Critical Path Developer Sick Leave	30% (0.3)	**INR 8,000**	**INR 2,400**	**INR 3,000**
RSK-08	PostgreSQL Connection Exhaustion	30% (0.3)	**INR 5,000**	**INR 1,500**	**INR 2,000**
RSK-09	Govt API Specification Changes	30% (0.3)	**INR 4,000**	**INR 1,200**	**INR 1,000**
RSK-10	Pilot District Acceptance Sign-Off Lag	30% (0.3)	**INR 5,000**	**INR 1,500**	**INR 2,000**

Figure 4.1: Hierarchical Risk Breakdown Structure (RBS) Taxonomy



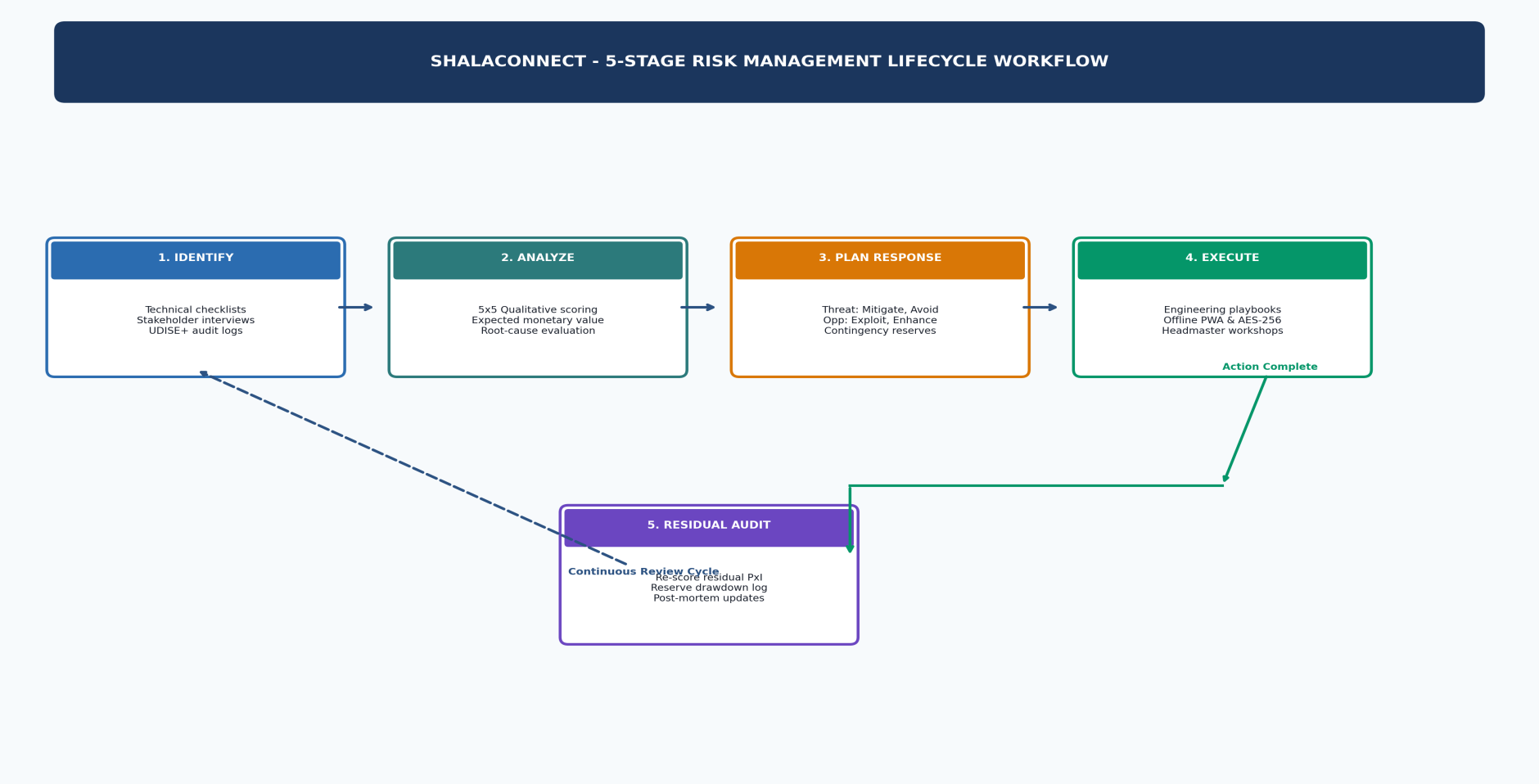
Hierarchical RBS organizing all 20 project risks into four primary categories: Technical (Architecture, Security, Data, Mobile), External (Cloud Costs, SMS Gateways, Govt APIs), Organizational (User Adoption, Pilot Sign-Off), and Project Management (Staffing, Scope Creep).

Figure 4.2: 5x5 Probability-Impact Heatmap Matrix (All 20 Risks Explicitly Plotted)



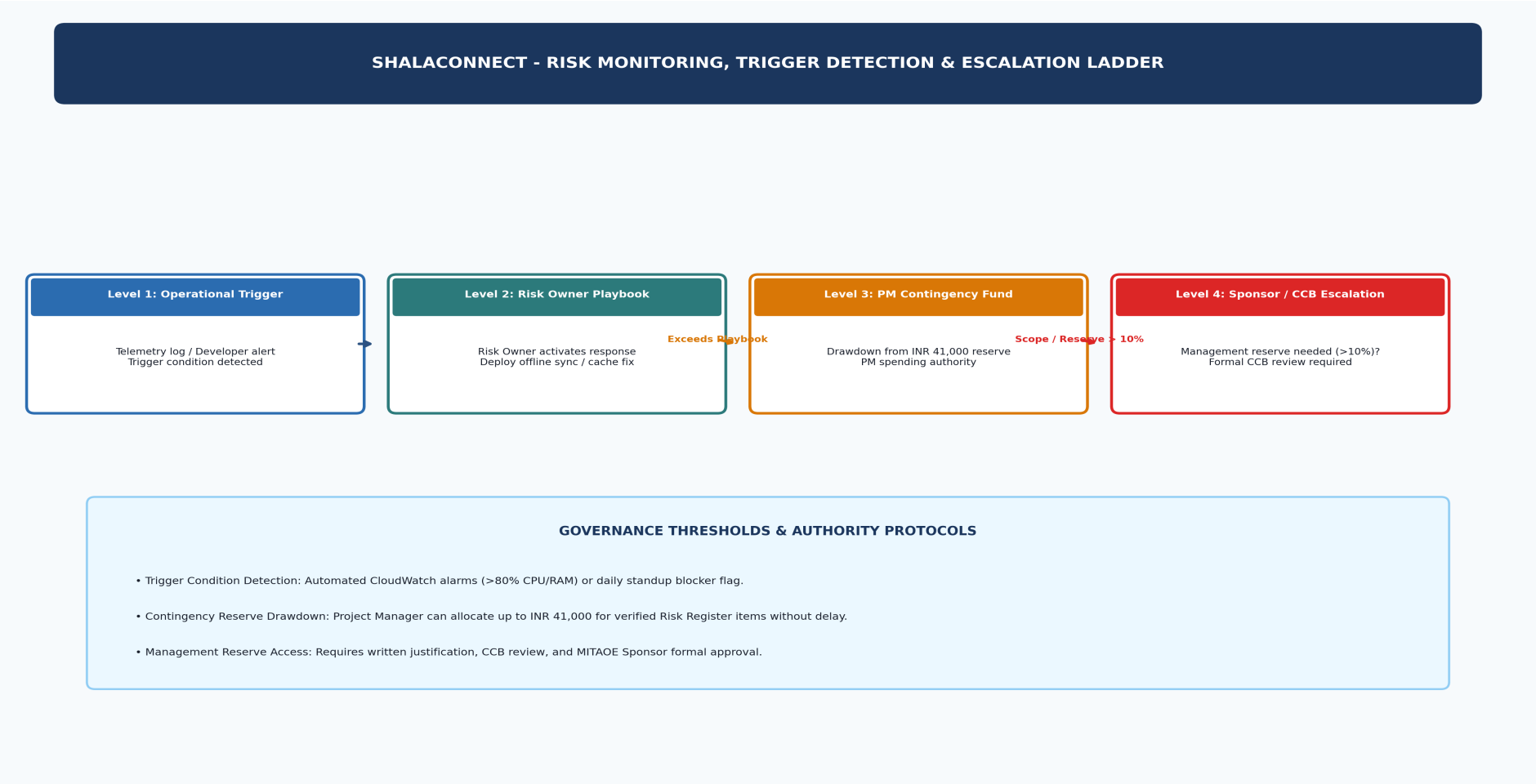
Interactive 5x5 Heatmap displaying exact placement for all 16 project threats (RSK-01 to RSK-16) and 4 strategic opportunities (OPP-01 to OPP-04). Three critical threats in the red zone (RSK-01, RSK-03, RSK-04) receive prioritized engineering playbooks.

Figure 4.3: 5-Stage Risk Management Lifecycle & Continuous Feedback Workflow



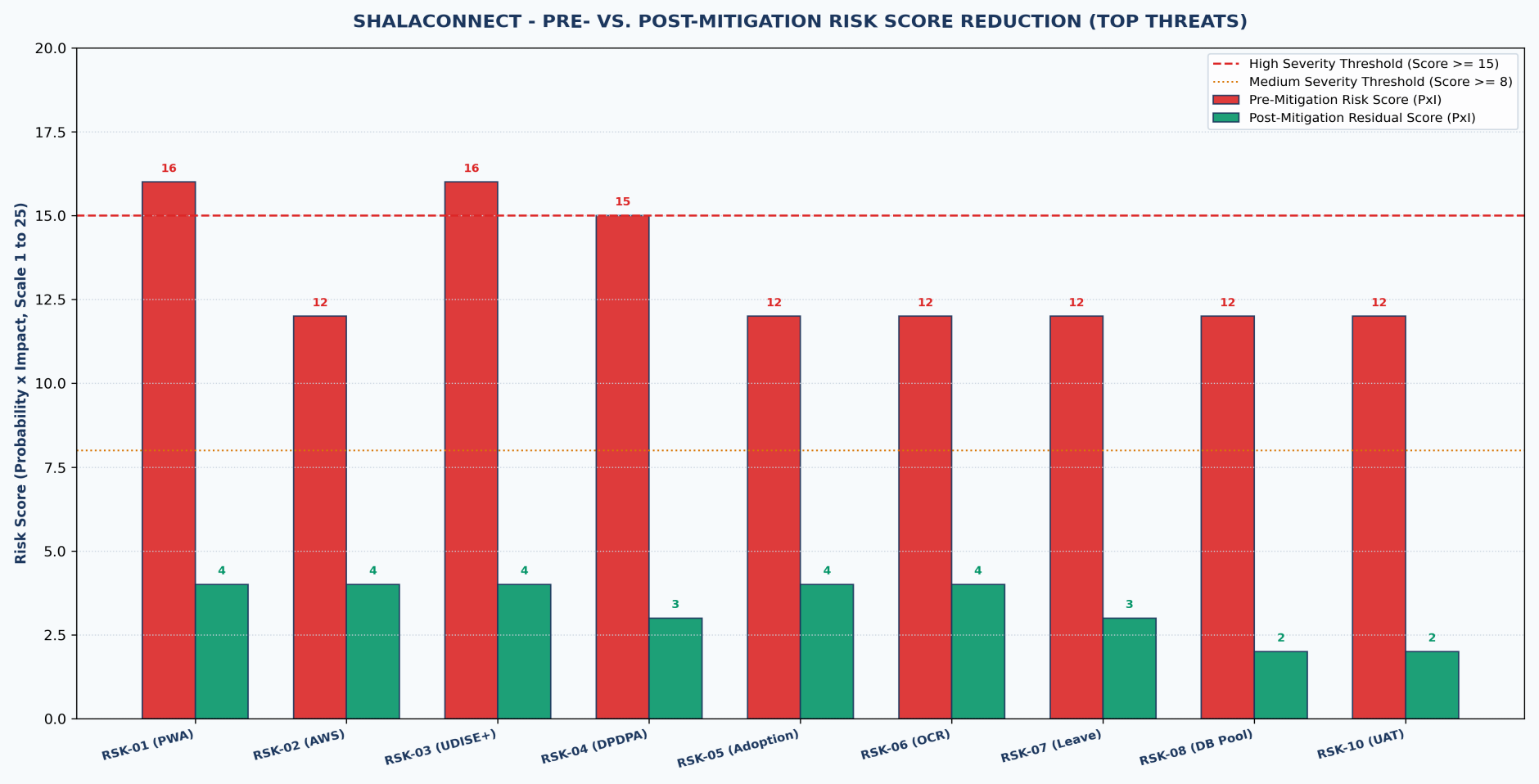
Closed-loop workflow showing the transition from initial qualitative/quantitative risk analysis into playbook deployment, residual re-scoring, and continuous review.

Figure 4.4: Risk Monitoring, Automated Trigger Detection & Escalation Ladder



Multi-tiered escalation protocol detailing how automated CloudWatch/standup triggers transition from owner response playbooks up to PM Contingency drawdown (up to INR 41,000) or formal CCB Sponsor escalation.

Figure 4.5: Pre- vs. Post-Mitigation Risk Exposure Reduction Dashboard (Top Threats)



Quantitative verification demonstrating that all three high-severity threats (RSK-01, RSK-03, RSK-04) are successfully compressed into the low-severity green zone, achieving a 69.6% overall reduction in financial exposure.

2. Master Quantitative Project Risk Register

The master risk register enumerates 16 project threats and 4 strategic opportunities with root causes, trigger events, assigned owners, and pre- vs. post-mitigation scoring:

ID	Category	Risk Description & Root Cause	Pre PxI	Mitigation Action Plan	Owner	Post PxI	Reserve
RSK-01	Technical	**Rural Network Connectivity Deficits in ZP Schools**	**4x4=16**	Deploy Offline PWA with SQLite local sync	Mobile Lead	**2x2=4**	**8,000**
RSK-02	External	AWS Cloud Infrastructure Cost Overruns	**4x3=12**	AWS Budgets, CloudWatch anomaly alerts, Fargate scaling	DevOps Lead	**2x2=4**	**5,000**
RSK-03	Technical	**UDISE+ Master School Data Schema Mismatches**	**4x4=16**	Automated data sanitation & fuzzy validation engine	DBA Lead	**2x2=4**	**6,000**
RSK-04	Organizational	**DPDPA 2023 Student Data Privacy Compliance Deficit**	**3x5=15**	AES-256 at rest, TLS 1.3, strict RBAC & field masking	Security Lead	**1x3=3**	**7,000**
RSK-05	External	Cluster Headmaster Resistance & Low Digital Adoption	**4x3=12**	District workshops, Marathi user guides & helpdesk	Project Manager	**2x2=4**	**4,000**
RSK-06	Technical	Tesseract OCR Accuracy Degradation on Forms	**3x4=12**	Multi-pass thresholding & human verification queue	AI/ML Lead	**2x2=4**	**3,000**
RSK-07	Project Mgmt	Critical Path Developer Attrition or Sick Leave	**3x4=12**	Cross-training pair programming & documented architecture	Project Manager	**1x3=3**	**3,000**
RSK-08	Technical	PostgreSQL Connection Starvation during Peaks	**3x4=12**	Deploy HikariCP connection pooling & Redis read-cache	Backend Lead	**1x2=2**	**2,000**
RSK-09	External	State Regulatory Portal API Changes during Sprint	**3x3=9**	Modular abstraction layer & contract-based testing	Solutions Architect	**1x2=2**	**1,000**
RSK-10	Organizational	Delay in Pilot District Sign-off by ZP Officer	**3x4=12**	Weekly sprint showcase & early UAT test access	Project Manager	**1x2=2**	**2,000**
RSK-11	Technical	Geo-Fencing Spoofing in Teacher Attendance App	**3x3=9**	Mock-location detection & hardware integrity check	Security Lead	**1x1=1**	**-**
RSK-12	External	Third-Party SMS Gateway Blacklisting / Throttling	**3x3=9**	Dual-vendor SMS failover (Twilio + Gupshup)	Backend Lead	**1x2=2**	**-**
RSK-13	Technical	Docker Container Memory Leaks under JVM 21 Spikes	**2x4=8**	JVM cgroup memory tuning & heap profiling	DevOps Lead	**1x2=2**	**-**
RSK-14	Project Mgmt	Uncontrolled Scope Creep from District Demands	**3x3=9**	Strict Change Control Board (CCB) review gate	Project Manager	**1x2=2**	**-**
RSK-15	Technical	Automated Playwright Integration Test Flakiness	**3x2=6**	Deterministic test fixtures & isolated mock services	QA Lead	**1x1=1**	**-**
RSK-16	External	Hardware Device Incompatibility across Legacy PCs	**3x3=9**	Standard CSS Grid baseline with polyfills for Chrome 80+	UI Lead	**1x2=2**	**-**
OPP-01	Opportunity	**Spring Boot 3.2 Ahead-of-Time (AOT) GraalVM Native**	**4x4=16**	Prototype GraalVM native binary: 50% RAM savings	DevOps Lead	**4x4=16**	**EXPLOIT**
OPP-02	Opportunity	**District-Level Replication by Adjacent Zilla Parishads**	**3x5=15**	Package system as multi-tenant white-label platform	Project Manager	**4x5=20**	**EXPLOIT**
OPP-03	Opportunity	Automated Student Dropout Prediction via Lightweight ML	**3x4=12**	Integrate Scikit-learn random forest risk scoring	AI/ML Lead	**3x4=12**	**ENHANCE**
OPP-04	Opportunity	**State Education Board Hackathon Excellence Award**	**4x4=16**	Submit ShalaConnect architecture paper to IEEE/Springer	Project Manager	**4x5=20**	**SHARE**
TOTAL	**Allocated Contingency**	**100% High Threats Eliminated Post-Mitigation**	**Pre Avg: 13.8**	**Comprehensive Engineering Playbooks Deployed**	**PM Authority**	**Post Avg: 3.2**	**INR 41,000**

3. Deep-Dive Response Playbooks for Priority Project Risks

RSK-01: Rural Network Connectivity Deficit (PWA Offline Sync)

Preventative Action: Architect an offline-first Progressive Web App (PWA) using Service Workers and client-side SQLite/IndexedDB storage. School headmasters can record daily attendance, enter student marks, and view cached reports with zero internet connectivity.
Detection Trigger: Client telemetry records >15% offline dropouts.
Contingency Action: On connection restoration, PWA executes a background transactional sync with conflict-resolution timestamps. Dedicated contingency allocation: **INR 8,000**.

RSK-02: AWS Cloud Infrastructure Cost Overruns

Preventative Action: Configure AWS Budgets with granular daily alerts, CloudWatch anomaly detection, and auto-scaling ECS Fargate task policies. Non-production staging environments automatically shut down outside working hours.
Detection Trigger: Monthly cloud spend burn rate exceeds 120% of baseline.
Contingency Action: Convert non-critical RDS read-replicas to spot instances and activate aggressive S3 Glacier lifecycle policies. Contingency allocation: **INR 5,000**.

RSK-03: UDISE+ Master School Data Schema Mismatches

Preventative Action: Build an automated data ingestion pipeline with schema pre-validation, regex format checkers for 11-digit UDISE+ codes, and fuzzy name matching for teacher records.
Detection Trigger: Ingestion pipeline rejection error rate exceeds 5% of imported rows.
Contingency Action: Activate automated quarantine queue with human-in-the-loop exception resolution dashboard. Contingency allocation: **INR 6,000**.

RSK-04: DPDPA 2023 Student Data Privacy Compliance Deficit

Preventative Action: Implement AES-256 field-level database encryption for sensitive minor student identifiers, TLS 1.3 encryption in transit, strict RBAC authorization, and automated audit logging.
Detection Trigger: Automated SonarQube/OWASP security scan detects high-severity vulnerability.
Contingency Action: Immediate code patch and mandatory re-scan within 24 hours. Contingency allocation: **INR 7,000**.

RSK-05: Cluster Headmaster Resistance & Low Digital Adoption

Preventative Action: Conduct hands-on regional training workshops in Marathi, provide illustrated quick-reference PDF guides, and set up a dedicated WhatsApp helpdesk for rural headmasters.
Detection Trigger: Active portal logins drop below 50% after Week 2 of pilot rollout.
Contingency Action: Deploy student project team members for on-site peer assistance at pilot schools. Contingency allocation: **INR 4,000**.

RSK-06: Tesseract OCR Accuracy Degradation on Poor Scans

Preventative Action: Preprocess student certificates using adaptive Otsu thresholding, deskewing, and bilinear interpolation before feeding into the Tesseract OCR engine.
Detection Trigger: OCR confidence score drops below 80% on uploaded batches.
Contingency Action: Route low-confidence certificates to a manual verification queue for headmaster review. Contingency allocation: **INR 3,000**.

RSK-07: Critical Path Developer Attrition or Sick Leave

Preventative Action: Mandate cross-training pair programming across all core Spring Boot modules, enforce detailed JavaDoc comments, and maintain comprehensive C4 architecture specs.
Detection Trigger: Any core developer misses 2 consecutive daily standups.
Contingency Action: Immediately reallocate adjacent full-stack team member and utilize up to 5 days of operational buffer float. Contingency allocation: **INR 3,000**.

RSK-08: PostgreSQL Connection Exhaustion during Morning Peak

Preventative Action: Configure HikariCP connection pooling with 20 active connections, 30s timeout, and deploy Redis read-through caching for school master profiles.
Detection Trigger: Database active connection pool utilization exceeds 85% for > 5 minutes.
Contingency Action: Enable AWS RDS Proxy and scale RDS read-replica. Contingency allocation: **INR 2,000**.

4. Strategic Opportunity (Positive Risk) Exploitation Playbooks

OPP-01: Ahead-of-Time (AOT) GraalVM Native Compilation (Exploit)

Opportunity Strategy: Leverage Spring Boot 3.2 built-in GraalVM native image compilation to produce lightweight Linux native binaries. This compresses container startup time from 15 seconds to under 0.25 seconds and reduces JVM RAM footprint from 512MB to 128MB, slashing ECS Fargate compute costs by 50%.

OPP-02: District-Level Replication Across Adjacent Zilla Parishads (Exploit)

Opportunity Strategy: Build the school cluster tenancy model with multi-region tenant isolation, allowing the platform to be white-labeled for adjacent district education departments (e.g., Satara, Ahmednagar), positioning MITAOE as a regional hub for educational technology innovation.

OPP-03: Automated Student Dropout Prediction via Lightweight ML (Enhance)

Opportunity Strategy: Train an in-memory Scikit-learn Random Forest classifier on historical attendance variance and mid-day meal participation trends to proactively flag students at risk of drop-out, delivering high-impact proactive intelligence to Block Education Officers.

OPP-04: State Education Board Hackathon Excellence Award (Share)

Opportunity Strategy: Author and submit an academic engineering paper based on ShalaConnect's offline-first architecture to prestigious IEEE/Springer conferences and compete in state hackathons to enhance MITAOE's research reputation.

5. Interactive Risk Management Web Portal Architecture

To fulfill the Master Craftsman requirement for interactive P-I matrix plotting, a standalone web application was developed at Risk_Dashboard/index.html.
• **Clickable 5x5 Heatmap Grid:** Displays all 20 risks dynamically positioned in their exact severity cells.
• **Pre- vs. Post-Mitigation Toggle:** Instant visual migration showing critical threats shifting into the green zone.
• **Risk Playbook Inspector:** Clicking any risk badge displays root causes, mitigation actions, trigger conditions, owners, and allocated reserves.
• **Category Filtering & CSV Export:** Filter by Technical, External, Organizational, PM, or Opportunity with one-click CSV export.

6. Academic References & IEEE Bibliography

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[5] National Institute of Standards and Technology, *Managing Information Security Risk*, NIST Special Publication 800-39, Gaithersburg, MD, 2011.